NAME:ANKIT KUMAR

ENROLLMENT: S25MCAG0010

LAB ASSIGNMENT: 8

Q1. Display all students along with the average marks of their department.

```
mysql> SELECT *,
    -> AVG(marks) OVER (PARTITION BY department) AS dept_avg
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+----------+
| student_id | student_name | department | subject | marks | dept_avg |
+------------+--------------+------------+---------+-------+----------+
|          1 | Amit         | CSE        | DBMS    |    78 |  80.7143 |
|          2 | Neha         | CSE        | DBMS    |    85 |  80.7143 |
|          3 | Raj          | CSE        | DBMS    |    85 |  80.7143 |
|          4 | Simran       | CSE        | OS      |    72 |  80.7143 |
|          5 | Karan        | CSE        | OS      |    88 |  80.7143 |
|         16 | Kavya        | CSE        | AI      |    90 |  80.7143 |
|         17 | Deepak       | CSE        | AI      |    67 |  80.7143 |
|          6 | Priya        | ECE        | DBMS    |    91 |  80.2857 |
|          7 | Vikas        | ECE        | DBMS    |    65 |  80.2857 |
|          8 | Riya         | ECE        | OS      |    74 |  80.2857 |
|          9 | Ankit        | ECE        | OS      |    74 |  80.2857 |
|         10 | Pooja        | ECE        | OS      |    82 |  80.2857 |
|         18 | Nisha        | ECE        | VLSI    |    88 |  80.2857 |
|         19 | Tarun        | ECE        | VLSI    |    88 |  80.2857 |
|         11 | Rahul        | ME         | Thermo  |    69 |  77.3333 |
|         12 | Sneha        | ME         | Thermo  |    73 |  77.3333 |
|         13 | Arjun        | ME         | Thermo  |    73 |  77.3333 |
|         14 | Meena        | ME         | Fluid   |    80 |  77.3333 |
|         15 | Rohit        | ME         | Fluid   |    92 |  77.3333 |
|         20 | Sonal        | ME         | Design  |    77 |  77.3333 |
+------------+--------------+------------+---------+-------+----------+
20 rows in set (0.03 sec)
```

Q2. Display each student along with the overall average marks.

```
mysql> SELECT *,
    -> AVG(marks) OVER () AS overall_avg
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+-------------+
| student_id | student_name | department | subject | marks | overall_avg |
+------------+--------------+------------+---------+-------+-------------+
|          1 | Amit         | CSE        | DBMS    |    78 |     79.5500 |
|          2 | Neha         | CSE        | DBMS    |    85 |     79.5500 |
|          3 | Raj          | CSE        | DBMS    |    85 |     79.5500 |
|          4 | Simran       | CSE        | OS      |    72 |     79.5500 |
|          5 | Karan        | CSE        | OS      |    88 |     79.5500 |
|          6 | Priya        | ECE        | DBMS    |    91 |     79.5500 |
|          7 | Vikas        | ECE        | DBMS    |    65 |     79.5500 |
|          8 | Riya         | ECE        | OS      |    74 |     79.5500 |
|          9 | Ankit        | ECE        | OS      |    74 |     79.5500 |
|         10 | Pooja        | ECE        | OS      |    82 |     79.5500 |
|         11 | Rahul        | ME         | Thermo  |    69 |     79.5500 |
|         12 | Sneha        | ME         | Thermo  |    73 |     79.5500 |
|         13 | Arjun        | ME         | Thermo  |    73 |     79.5500 |
|         14 | Meena        | ME         | Fluid   |    80 |     79.5500 |
|         15 | Rohit        | ME         | Fluid   |    92 |     79.5500 |
|         16 | Kavya        | CSE        | AI      |    90 |     79.5500 |
|         17 | Deepak       | CSE        | AI      |    67 |     79.5500 |
|         18 | Nisha        | ECE        | VLSI    |    88 |     79.5500 |
|         19 | Tarun        | ECE        | VLSI    |    88 |     79.5500 |
|         20 | Sonal        | ME         | Design  |    77 |     79.5500 |
+------------+--------------+------------+---------+-------+-------------+
20 rows in set (0.00 sec)
```

Q3. Assign row number to students within each subject based on marks (highest first).

```
mysql> SELECT *,
    -> ROW_NUMBER() OVER (PARTITION BY subject ORDER BY marks DESC) AS row_num
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+---------+
| student_id | student_name | department | subject | marks | row_num |
+------------+--------------+------------+---------+-------+---------+
|         16 | Kavya        | CSE        | AI      |    90 |       1 |
|         17 | Deepak       | CSE        | AI      |    67 |       2 |
|          6 | Priya        | ECE        | DBMS    |    91 |       1 |
|          2 | Neha         | CSE        | DBMS    |    85 |       2 |
|          3 | Raj          | CSE        | DBMS    |    85 |       3 |
|          1 | Amit         | CSE        | DBMS    |    78 |       4 |
|          7 | Vikas        | ECE        | DBMS    |    65 |       5 |
|         20 | Sonal        | ME         | Design  |    77 |       1 |
|         15 | Rohit        | ME         | Fluid   |    92 |       1 |
|         14 | Meena        | ME         | Fluid   |    80 |       2 |
|          5 | Karan        | CSE        | OS      |    88 |       1 |
|         10 | Pooja        | ECE        | OS      |    82 |       2 |
|          8 | Riya         | ECE        | OS      |    74 |       3 |
|          9 | Ankit        | ECE        | OS      |    74 |       4 |
|          4 | Simran       | CSE        | OS      |    72 |       5 |
|         12 | Sneha        | ME         | Thermo  |    73 |       1 |
|         13 | Arjun        | ME         | Thermo  |    73 |       2 |
|         11 | Rahul        | ME         | Thermo  |    69 |       3 |
|         18 | Nisha        | ECE        | VLSI    |    88 |       1 |
|         19 | Tarun        | ECE        | VLSI    |    88 |       2 |
+------------+--------------+------------+---------+-------+---------+
20 rows in set (0.01 sec)
```

Q4. Find rank of students within each subject.

```
mysql> SELECT *,
    -> RANK() OVER (PARTITION BY subject ORDER BY marks DESC) AS rnk
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+-----+
| student_id | student_name | department | subject | marks | rnk |
+------------+--------------+------------+---------+-------+-----+
|         16 | Kavya        | CSE        | AI      |    90 |   1 |
|         17 | Deepak       | CSE        | AI      |    67 |   2 |
|          6 | Priya        | ECE        | DBMS    |    91 |   1 |
|          2 | Neha         | CSE        | DBMS    |    85 |   2 |
|          3 | Raj          | CSE        | DBMS    |    85 |   2 |
|          1 | Amit         | CSE        | DBMS    |    78 |   4 |
|          7 | Vikas        | ECE        | DBMS    |    65 |   5 |
|         20 | Sonal        | ME         | Design  |    77 |   1 |
|         15 | Rohit        | ME         | Fluid   |    92 |   1 |
|         14 | Meena        | ME         | Fluid   |    80 |   2 |
|          5 | Karan        | CSE        | OS      |    88 |   1 |
|         10 | Pooja        | ECE        | OS      |    82 |   2 |
|          8 | Riya         | ECE        | OS      |    74 |   3 |
|          9 | Ankit        | ECE        | OS      |    74 |   3 |
|          4 | Simran       | CSE        | OS      |    72 |   5 |
|         12 | Sneha        | ME         | Thermo  |    73 |   1 |
|         13 | Arjun        | ME         | Thermo  |    73 |   1 |
|         11 | Rahul        | ME         | Thermo  |    69 |   3 |
|         18 | Nisha        | ECE        | VLSI    |    88 |   1 |
|         19 | Tarun        | ECE        | VLSI    |    88 |   1 |
+------------+--------------+------------+---------+-------+-----+
20 rows in set (0.00 sec)
```

Q5. Find dense rank of students within each subject.

```
mysql> SELECT *, DENSE_RANK() OVER (PARTITION BY subject ORDER BY marks DESC) AS dr FROM StudentMarks;
+------------+--------------+------------+---------+-------+----+
| student_id | student_name | department | subject | marks | dr |
+------------+--------------+------------+---------+-------+----+
|         16 | Kavya        | CSE        | AI      |    90 |  1 |
|         17 | Deepak       | CSE        | AI      |    67 |  2 |
|          6 | Priya        | ECE        | DBMS    |    91 |  1 |
|          2 | Neha         | CSE        | DBMS    |    85 |  2 |
|          3 | Raj          | CSE        | DBMS    |    85 |  2 |
|          1 | Amit         | CSE        | DBMS    |    78 |  3 |
|          7 | Vikas        | ECE        | DBMS    |    65 |  4 |
|         20 | Sonal        | ME         | Design  |    77 |  1 |
|         15 | Rohit        | ME         | Fluid   |    92 |  1 |
|         14 | Meena        | ME         | Fluid   |    80 |  2 |
|          5 | Karan        | CSE        | OS      |    88 |  1 |
|         10 | Pooja        | ECE        | OS      |    82 |  2 |
|          8 | Riya         | ECE        | OS      |    74 |  3 |
|          9 | Ankit        | ECE        | OS      |    74 |  3 |
|          4 | Simran       | CSE        | OS      |    72 |  4 |
|         12 | Sneha        | ME         | Thermo  |    73 |  1 |
|         13 | Arjun        | ME         | Thermo  |    73 |  1 |
|         11 | Rahul        | ME         | Thermo  |    69 |  2 |
|         18 | Nisha        | ECE        | VLSI    |    88 |  1 |
|         19 | Tarun        | ECE        | VLSI    |    88 |  1 |
+------------+--------------+------------+---------+-------+----+
20 rows in set (0.00 sec)
```

Q6. Display each student with maximum marks in their subject.

```
mysql> SELECT *,
    ->        MAX(marks) OVER(PARTITION BY subject) AS max_marks
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+-----------+
| student_id | student_name | department | subject | marks | max_marks |
+------------+--------------+------------+---------+-------+-----------+
|         16 | Kavya        | CSE        | AI      |    90 |        90 |
|         17 | Deepak       | CSE        | AI      |    67 |        90 |
|          1 | Amit         | CSE        | DBMS    |    78 |        91 |
|          2 | Neha         | CSE        | DBMS    |    85 |        91 |
|          3 | Raj          | CSE        | DBMS    |    85 |        91 |
|          6 | Priya        | ECE        | DBMS    |    91 |        91 |
|          7 | Vikas        | ECE        | DBMS    |    65 |        91 |
|         20 | Sonal        | ME         | Design  |    77 |        77 |
|         14 | Meena        | ME         | Fluid   |    80 |        92 |
|         15 | Rohit        | ME         | Fluid   |    92 |        92 |
|          4 | Simran       | CSE        | OS      |    72 |        88 |
|          5 | Karan        | CSE        | OS      |    88 |        88 |
|          8 | Riya         | ECE        | OS      |    74 |        88 |
|          9 | Ankit        | ECE        | OS      |    74 |        88 |
|         10 | Pooja        | ECE        | OS      |    82 |        88 |
|         11 | Rahul        | ME         | Thermo  |    69 |        73 |
|         12 | Sneha        | ME         | Thermo  |    73 |        73 |
|         13 | Arjun        | ME         | Thermo  |    73 |        73 |
|         18 | Nisha        | ECE        | VLSI    |    88 |        88 |
|         19 | Tarun        | ECE        | VLSI    |    88 |        88 |
+------------+--------------+------------+---------+-------+-----------+
20 rows in set (0.08 sec)
```

Q7. Find students who scored above the average marks of their subject.

```
mysql> SELECT *
    -> FROM (
    ->     SELECT *,
    ->            AVG(marks) OVER(PARTITION BY subject) AS avg_marks
    ->     FROM StudentMarks
    -> ) t
    -> WHERE marks > avg_marks;
+------------+--------------+------------+---------+-------+-----------+
| student_id | student_name | department | subject | marks | avg_marks |
+------------+--------------+------------+---------+-------+-----------+
|         16 | Kavya        | CSE        | AI      |    90 |   78.5000 |
|          2 | Neha         | CSE        | DBMS    |    85 |   80.8000 |
|          3 | Raj          | CSE        | DBMS    |    85 |   80.8000 |
|          6 | Priya        | ECE        | DBMS    |    91 |   80.8000 |
|         15 | Rohit        | ME         | Fluid   |    92 |   86.0000 |
|          5 | Karan        | CSE        | OS      |    88 |   78.0000 |
|         10 | Pooja        | ECE        | OS      |    82 |   78.0000 |
|         12 | Sneha        | ME         | Thermo  |    73 |   71.6667 |
|         13 | Arjun        | ME         | Thermo  |    73 |   71.6667 |
+------------+--------------+------------+---------+-------+-----------+
9 rows in set (0.01 sec)
```

Q8. Find top 2 students in each subject.

```
mysql> SELECT *
    -> FROM (
    ->     SELECT *,
    ->            RANK() OVER(PARTITION BY subject ORDER BY marks DESC) AS rnk
    ->     FROM StudentMarks
    -> ) t
    -> WHERE rnk <= 2;
+------------+--------------+------------+---------+-------+-----+
| student_id | student_name | department | subject | marks | rnk |
+------------+--------------+------------+---------+-------+-----+
|         16 | Kavya        | CSE        | AI      |    90 |   1 |
|         17 | Deepak       | CSE        | AI      |    67 |   2 |
|          6 | Priya        | ECE        | DBMS    |    91 |   1 |
|          2 | Neha         | CSE        | DBMS    |    85 |   2 |
|          3 | Raj          | CSE        | DBMS    |    85 |   2 |
|         20 | Sonal        | ME         | Design  |    77 |   1 |
|         15 | Rohit        | ME         | Fluid   |    92 |   1 |
|         14 | Meena        | ME         | Fluid   |    80 |   2 |
|          5 | Karan        | CSE        | OS      |    88 |   1 |
|         10 | Pooja        | ECE        | OS      |    82 |   2 |
|         12 | Sneha        | ME         | Thermo  |    73 |   1 |
|         13 | Arjun        | ME         | Thermo  |    73 |   1 |
|         18 | Nisha        | ECE        | VLSI    |    88 |   1 |
|         19 | Tarun        | ECE        | VLSI    |    88 |   1 |
+------------+--------------+------------+---------+-------+-----+
14 rows in set (0.01 sec)
```

Q9. Find second highest marks in each subject.

```
mysql> SELECT *
    -> FROM (
    ->     SELECT *,
    ->            DENSE_RANK() OVER(PARTITION BY subject ORDER BY marks DESC) AS dr
    ->     FROM StudentMarks
    -> ) t
    -> WHERE dr = 2;
+------------+--------------+------------+---------+-------+----+
| student_id | student_name | department | subject | marks | dr |
+------------+--------------+------------+---------+-------+----+
|         17 | Deepak       | CSE        | AI      |    67 |  2 |
|          2 | Neha         | CSE        | DBMS    |    85 |  2 |
|          3 | Raj          | CSE        | DBMS    |    85 |  2 |
|         14 | Meena        | ME         | Fluid   |    80 |  2 |
|         10 | Pooja        | ECE        | OS      |    82 |  2 |
|         11 | Rahul        | ME         | Thermo  |    69 |  2 |
+------------+--------------+------------+---------+-------+----+
6 rows in set (0.00 sec)
```

Q10. Display students along with running total of marks within each department.

```
mysql> SELECT *,
    ->        SUM(marks) OVER(PARTITION BY department ORDER BY student_id) AS running_total
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+---------------+
| student_id | student_name | department | subject | marks | running_total |
+------------+--------------+------------+---------+-------+---------------+
|          1 | Amit         | CSE        | DBMS    |    78 |            78 |
|          2 | Neha         | CSE        | DBMS    |    85 |           163 |
|          3 | Raj          | CSE        | DBMS    |    85 |           248 |
|          4 | Simran       | CSE        | OS      |    72 |           320 |
|          5 | Karan        | CSE        | OS      |    88 |           408 |
|         16 | Kavya        | CSE        | AI      |    90 |           498 |
|         17 | Deepak       | CSE        | AI      |    67 |           565 |
|          6 | Priya        | ECE        | DBMS    |    91 |            91 |
|          7 | Vikas        | ECE        | DBMS    |    65 |           156 |
|          8 | Riya         | ECE        | OS      |    74 |           230 |
|          9 | Ankit        | ECE        | OS      |    74 |           304 |
|         10 | Pooja        | ECE        | OS      |    82 |           386 |
|         18 | Nisha        | ECE        | VLSI    |    88 |           474 |
|         19 | Tarun        | ECE        | VLSI    |    88 |           562 |
|         11 | Rahul        | ME         | Thermo  |    69 |            69 |
|         12 | Sneha        | ME         | Thermo  |    73 |           142 |
|         13 | Arjun        | ME         | Thermo  |    73 |           215 |
|         14 | Meena        | ME         | Fluid   |    80 |           295 |
|         15 | Rohit        | ME         | Fluid   |    92 |           387 |
|         20 | Sonal        | ME         | Design  |    77 |           464 |
+------------+--------------+------------+---------+-------+---------------+
20 rows in set (0.00 sec)
```

Q11. Display students along with previous student’s marks (ordered by student_id).

```
mysql> SELECT *,
    ->        LAG(marks) OVER(ORDER BY student_id) AS prev_marks
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+------------+
| student_id | student_name | department | subject | marks | prev_marks |
+------------+--------------+------------+---------+-------+------------+
|          1 | Amit         | CSE        | DBMS    |    78 |       NULL |
|          2 | Neha         | CSE        | DBMS    |    85 |         78 |
|          3 | Raj          | CSE        | DBMS    |    85 |         85 |
|          4 | Simran       | CSE        | OS      |    72 |         85 |
|          5 | Karan        | CSE        | OS      |    88 |         72 |
|          6 | Priya        | ECE        | DBMS    |    91 |         88 |
|          7 | Vikas        | ECE        | DBMS    |    65 |         91 |
|          8 | Riya         | ECE        | OS      |    74 |         65 |
|          9 | Ankit        | ECE        | OS      |    74 |         74 |
|         10 | Pooja        | ECE        | OS      |    82 |         74 |
|         11 | Rahul        | ME         | Thermo  |    69 |         82 |
|         12 | Sneha        | ME         | Thermo  |    73 |         69 |
|         13 | Arjun        | ME         | Thermo  |    73 |         73 |
|         14 | Meena        | ME         | Fluid   |    80 |         73 |
|         15 | Rohit        | ME         | Fluid   |    92 |         80 |
|         16 | Kavya        | CSE        | AI      |    90 |         92 |
|         17 | Deepak       | CSE        | AI      |    67 |         90 |
|         18 | Nisha        | ECE        | VLSI    |    88 |         67 |
|         19 | Tarun        | ECE        | VLSI    |    88 |         88 |
|         20 | Sonal        | ME         | Design  |    77 |         88 |
+------------+--------------+------------+---------+-------+------------+
20 rows in set (0.00 sec)
```

Q12. Display students along with next student’s marks.

```
mysql> SELECT *,
    ->        LEAD(marks) OVER(ORDER BY student_id) AS next_marks
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+------------+
| student_id | student_name | department | subject | marks | next_marks |
+------------+--------------+------------+---------+-------+------------+
|          1 | Amit         | CSE        | DBMS    |    78 |         85 |
|          2 | Neha         | CSE        | DBMS    |    85 |         85 |
|          3 | Raj          | CSE        | DBMS    |    85 |         72 |
|          4 | Simran       | CSE        | OS      |    72 |         88 |
|          5 | Karan        | CSE        | OS      |    88 |         91 |
|          6 | Priya        | ECE        | DBMS    |    91 |         65 |
|          7 | Vikas        | ECE        | DBMS    |    65 |         74 |
|          8 | Riya         | ECE        | OS      |    74 |         74 |
|          9 | Ankit        | ECE        | OS      |    74 |         82 |
|         10 | Pooja        | ECE        | OS      |    82 |         69 |
|         11 | Rahul        | ME         | Thermo  |    69 |         73 |
|         12 | Sneha        | ME         | Thermo  |    73 |         73 |
|         13 | Arjun        | ME         | Thermo  |    73 |         80 |
|         14 | Meena        | ME         | Fluid   |    80 |         92 |
|         15 | Rohit        | ME         | Fluid   |    92 |         90 |
|         16 | Kavya        | CSE        | AI      |    90 |         67 |
|         17 | Deepak       | CSE        | AI      |    67 |         88 |
|         18 | Nisha        | ECE        | VLSI    |    88 |         88 |
|         19 | Tarun        | ECE        | VLSI    |    88 |         77 |
|         20 | Sonal        | ME         | Design  |    77 |       NULL |
+------------+--------------+------------+---------+-------+------------+
20 rows in set (0.00 sec)
```

Q13. Find students whose marks are equal to the previous student's marks.

```
mysql> SELECT *
    -> FROM (
    ->     SELECT *,
    ->            LAG(marks) OVER(ORDER BY student_id) AS prev_marks
    ->     FROM StudentMarks
    -> ) t
    -> WHERE marks = prev_marks;
+------------+--------------+------------+---------+-------+------------+
| student_id | student_name | department | subject | marks | prev_marks |
+------------+--------------+------------+---------+-------+------------+
|          3 | Raj          | CSE        | DBMS    |    85 |         85 |
|          9 | Ankit        | ECE        | OS      |    74 |         74 |
|         13 | Arjun        | ME         | Thermo  |    73 |         73 |
|         19 | Tarun        | ECE        | VLSI    |    88 |         88 |
+------------+--------------+------------+---------+-------+------------+
4 rows in set (0.00 sec)
```

Q14. Find students whose marks are greater than previous student's marks.

```
mysql> SELECT *
    -> FROM (
    ->     SELECT *,
    ->            LAG(marks) OVER(ORDER BY student_id) AS prev_marks
    ->     FROM StudentMarks
    -> ) t
    -> WHERE marks > prev_marks;
+------------+--------------+------------+---------+-------+------------+
| student_id | student_name | department | subject | marks | prev_marks |
+------------+--------------+------------+---------+-------+------------+
|          2 | Neha         | CSE        | DBMS    |    85 |         78 |
|          5 | Karan        | CSE        | OS      |    88 |         72 |
|          6 | Priya        | ECE        | DBMS    |    91 |         88 |
|          8 | Riya         | ECE        | OS      |    74 |         65 |
|         10 | Pooja        | ECE        | OS      |    82 |         74 |
|         12 | Sneha        | ME         | Thermo  |    73 |         69 |
|         14 | Meena        | ME         | Fluid   |    80 |         73 |
|         15 | Rohit        | ME         | Fluid   |    92 |         80 |
|         18 | Nisha        | ECE        | VLSI    |    88 |         67 |
+------------+--------------+------------+---------+-------+------------+
9 rows in set (0.00 sec)
```

Q15. Divide students into 4 groups (quartiles) based on marks.

```
mysql> SELECT *,
    ->        NTILE(4) OVER(ORDER BY marks DESC) AS quartile
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+----------+
| student_id | student_name | department | subject | marks | quartile |
+------------+--------------+------------+---------+-------+----------+
|         15 | Rohit        | ME         | Fluid   |    92 |        1 |
|          6 | Priya        | ECE        | DBMS    |    91 |        1 |
|         16 | Kavya        | CSE        | AI      |    90 |        1 |
|          5 | Karan        | CSE        | OS      |    88 |        1 |
|         18 | Nisha        | ECE        | VLSI    |    88 |        1 |
|         19 | Tarun        | ECE        | VLSI    |    88 |        2 |
|          2 | Neha         | CSE        | DBMS    |    85 |        2 |
|          3 | Raj          | CSE        | DBMS    |    85 |        2 |
|         10 | Pooja        | ECE        | OS      |    82 |        2 |
|         14 | Meena        | ME         | Fluid   |    80 |        2 |
|          1 | Amit         | CSE        | DBMS    |    78 |        3 |
|         20 | Sonal        | ME         | Design  |    77 |        3 |
|          8 | Riya         | ECE        | OS      |    74 |        3 |
|          9 | Ankit        | ECE        | OS      |    74 |        3 |
|         12 | Sneha        | ME         | Thermo  |    73 |        3 |
|         13 | Arjun        | ME         | Thermo  |    73 |        4 |
|          4 | Simran       | CSE        | OS      |    72 |        4 |
|         11 | Rahul        | ME         | Thermo  |    69 |        4 |
|         17 | Deepak       | CSE        | AI      |    67 |        4 |
|          7 | Vikas        | ECE        | DBMS    |    65 |        4 |
+------------+--------------+------------+---------+-------+----------+
20 rows in set (0.00 sec)
```

Q16. Find students who are in the top 50% within their subject.

```
mysql> SELECT *
    -> FROM (
    ->     SELECT *,
    ->            NTILE(2) OVER(PARTITION BY subject ORDER BY marks DESC) AS grp
    ->     FROM StudentMarks
    -> ) t
    -> WHERE grp = 1;
+------------+--------------+------------+---------+-------+-----+
| student_id | student_name | department | subject | marks | grp |
+------------+--------------+------------+---------+-------+-----+
|         16 | Kavya        | CSE        | AI      |    90 |   1 |
|          6 | Priya        | ECE        | DBMS    |    91 |   1 |
|          2 | Neha         | CSE        | DBMS    |    85 |   1 |
|          3 | Raj          | CSE        | DBMS    |    85 |   1 |
|         20 | Sonal        | ME         | Design  |    77 |   1 |
|         15 | Rohit        | ME         | Fluid   |    92 |   1 |
|          5 | Karan        | CSE        | OS      |    88 |   1 |
|         10 | Pooja        | ECE        | OS      |    82 |   1 |
|          8 | Riya         | ECE        | OS      |    74 |   1 |
|         12 | Sneha        | ME         | Thermo  |    73 |   1 |
|         13 | Arjun        | ME         | Thermo  |    73 |   1 |
|         18 | Nisha        | ECE        | VLSI    |    88 |   1 |
+------------+--------------+------------+---------+-------+-----+
12 rows in set (0.00 sec)
```

Q17. Calculate moving average of marks (current + previous row).

```
mysql> SELECT *,
    ->        AVG(marks) OVER(ORDER BY student_id ROWS BETWEEN 1 PRECEDING AND CURRENT ROW) AS moving_avg
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+------------+
| student_id | student_name | department | subject | marks | moving_avg |
+------------+--------------+------------+---------+-------+------------+
|          1 | Amit         | CSE        | DBMS    |    78 |    78.0000 |
|          2 | Neha         | CSE        | DBMS    |    85 |    81.5000 |
|          3 | Raj          | CSE        | DBMS    |    85 |    85.0000 |
|          4 | Simran       | CSE        | OS      |    72 |    78.5000 |
|          5 | Karan        | CSE        | OS      |    88 |    80.0000 |
|          6 | Priya        | ECE        | DBMS    |    91 |    89.5000 |
|          7 | Vikas        | ECE        | DBMS    |    65 |    78.0000 |
|          8 | Riya         | ECE        | OS      |    74 |    69.5000 |
|          9 | Ankit        | ECE        | OS      |    74 |    74.0000 |
|         10 | Pooja        | ECE        | OS      |    82 |    78.0000 |
|         11 | Rahul        | ME         | Thermo  |    69 |    75.5000 |
|         12 | Sneha        | ME         | Thermo  |    73 |    71.0000 |
|         13 | Arjun        | ME         | Thermo  |    73 |    73.0000 |
|         14 | Meena        | ME         | Fluid   |    80 |    76.5000 |
|         15 | Rohit        | ME         | Fluid   |    92 |    86.0000 |
|         16 | Kavya        | CSE        | AI      |    90 |    91.0000 |
|         17 | Deepak       | CSE        | AI      |    67 |    78.5000 |
|         18 | Nisha        | ECE        | VLSI    |    88 |    77.5000 |
|         19 | Tarun        | ECE        | VLSI    |    88 |    88.0000 |
|         20 | Sonal        | ME         | Design  |    77 |    82.5000 |
+------------+--------------+------------+---------+-------+------------+
20 rows in set (0.00 sec)
```

Q18. Display:

- student marks
- subject average

- difference from subject average

```
mysql> SELECT *,
    ->        AVG(marks) OVER(PARTITION BY subject) AS subject_avg,
    ->        marks - AVG(marks) OVER(PARTITION BY subject) AS diff
    -> FROM StudentMarks;
+------------+--------------+------------+---------+-------+-------------+----------+
| student_id | student_name | department | subject | marks | subject_avg | diff     |
+------------+--------------+------------+---------+-------+-------------+----------+
|         16 | Kavya        | CSE        | AI      |    90 |     78.5000 |  11.5000 |
|         17 | Deepak       | CSE        | AI      |    67 |     78.5000 | -11.5000 |
|          1 | Amit         | CSE        | DBMS    |    78 |     80.8000 |  -2.8000 |
|          2 | Neha         | CSE        | DBMS    |    85 |     80.8000 |   4.2000 |
|          3 | Raj          | CSE        | DBMS    |    85 |     80.8000 |   4.2000 |
|          6 | Priya        | ECE        | DBMS    |    91 |     80.8000 |  10.2000 |
|          7 | Vikas        | ECE        | DBMS    |    65 |     80.8000 | -15.8000 |
|         20 | Sonal        | ME         | Design  |    77 |     77.0000 |   0.0000 |
|         14 | Meena        | ME         | Fluid   |    80 |     86.0000 |  -6.0000 |
|         15 | Rohit        | ME         | Fluid   |    92 |     86.0000 |   6.0000 |
|          4 | Simran       | CSE        | OS      |    72 |     78.0000 |  -6.0000 |
|          5 | Karan        | CSE        | OS      |    88 |     78.0000 |  10.0000 |
|          8 | Riya         | ECE        | OS      |    74 |     78.0000 |  -4.0000 |
|          9 | Ankit        | ECE        | OS      |    74 |     78.0000 |  -4.0000 |
|         10 | Pooja        | ECE        | OS      |    82 |     78.0000 |   4.0000 |
|         11 | Rahul        | ME         | Thermo  |    69 |     71.6667 |  -2.6667 |
|         12 | Sneha        | ME         | Thermo  |    73 |     71.6667 |   1.3333 |
|         13 | Arjun        | ME         | Thermo  |    73 |     71.6667 |   1.3333 |
|         18 | Nisha        | ECE        | VLSI    |    88 |     88.0000 |   0.0000 |
|         19 | Tarun        | ECE        | VLSI    |    88 |     88.0000 |   0.0000 |
+------------+--------------+------------+---------+-------+-------------+----------+
20 rows in set (0.01 sec)
```

Q19. Find students whose marks are equal to subject maximum but not unique.

```
mysql> SELECT *
    -> FROM (
    ->     SELECT *,
    ->            MAX(marks) OVER(PARTITION BY subject) AS max_marks,
    ->            COUNT(*) OVER(PARTITION BY subject, marks) AS cnt
    ->     FROM StudentMarks
    -> ) t
    -> WHERE marks = max_marks AND cnt > 1;
+------------+--------------+------------+---------+-------+-----------+-----+
| student_id | student_name | department | subject | marks | max_marks | cnt |
+------------+--------------+------------+---------+-------+-----------+-----+
|         12 | Sneha        | ME         | Thermo  |    73 |        73 |   2 |
|         13 | Arjun        | ME         | Thermo  |    73 |        73 |   2 |
|         18 | Nisha        | ECE        | VLSI    |    88 |        88 |   2 |
|         19 | Tarun        | ECE        | VLSI    |    88 |        88 |   2 |
+------------+--------------+------------+---------+-------+-----------+-----+
4 rows in set (0.00 sec)
```

Q20. Find students whose marks are:

- in top 2 of their subject
- but not in top 3 overall

```
mysql> SELECT *
    -> FROM (
    ->     SELECT *,
    ->            RANK() OVER(PARTITION BY subject ORDER BY marks DESC) AS sub_rank,
    ->            RANK() OVER(ORDER BY marks DESC) AS overall_rank
    ->     FROM StudentMarks
    -> ) t
    -> WHERE sub_rank <= 2 AND overall_rank > 3;
+------------+--------------+------------+---------+-------+----------+--------------+
| student_id | student_name | department | subject | marks | sub_rank | overall_rank |
+------------+--------------+------------+---------+-------+----------+--------------+
|          5 | Karan        | CSE        | OS      |    88 |        1 |            4 |
|         18 | Nisha        | ECE        | VLSI    |    88 |        1 |            4 |
|         19 | Tarun        | ECE        | VLSI    |    88 |        1 |            4 |
|          2 | Neha         | CSE        | DBMS    |    85 |        2 |            7 |
|          3 | Raj          | CSE        | DBMS    |    85 |        2 |            7 |
|         10 | Pooja        | ECE        | OS      |    82 |        2 |            9 |
|         14 | Meena        | ME         | Fluid   |    80 |        2 |           10 |
|         20 | Sonal        | ME         | Design  |    77 |        1 |           12 |
|         12 | Sneha        | ME         | Thermo  |    73 |        1 |           15 |
|         13 | Arjun        | ME         | Thermo  |    73 |        1 |           15 |
|         17 | Deepak       | CSE        | AI      |    67 |        2 |           19 |
+------------+--------------+------------+---------+-------+----------+--------------+
11 rows in set (0.00 sec)
```